Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

Exercise 7.1 · Constructing a line

The plane is equipped with an orthonormal frame. The line d passes through $A(-1,3)$ and has the vector $\vec{n} = (2,5)$ as a normal vector.

1. Write the condition for a point $M(x,y)$ to belong to d using a scalar product. Deduce a Cartesian equation of d.
2. Give a nonzero direction vector of d, then the coordinates of a point of d other than A.
3. Check your point by substitution in the equation.

Exercise 7.2 · Calculating a projection

In an orthonormal frame, given the point $P(4,3)$ and the line d with equation $2x - y = 0$. Let H be the orthogonal projection of P onto d.

1. Determine the coordinates of H by translating the two conditions: H belongs to d and $\overrightarrow{PH}$ is normal to d.
2. Check these two conditions separately with the coordinates obtained.
3. Calculate the distance from P to the line d in exact form.

Exercise 7.3 · Recognising a circle

Consider the set of points $M(x,y)$ of the orthonormal plane whose coordinates satisfy

$$x^2 + y^2 - 6x + 4y - 12 = 0.$$

1. Show that this set is a circle and state its centre and radius.
2. Determine all its points of intersection with the line with equation $y = -2$.
3. Interpret geometrically the position of these two points on the circle.

Exercise 7.4 · Perpendicular bisector

In an orthonormal frame, given $A(-1,1)$ and $B(3,3)$. We look for an equation of the perpendicular bisector of $[AB]$.

1. Determine this equation using the geometric properties of the perpendicular bisector. Explain the choice of a point and of a normal vector.
2. Check that the midpoint of $[AB]$ satisfies the equation obtained.

Exercise 7.5 · The best access

A straight road is represented by the line d with equation $y = 2x$ in an orthonormal frame whose unit of length is 1 km. A building is located at the point $P(4, 3)$. We want to build a straight access path joining the building to a point on the road.

1. Determine the point of the road that makes this access as short as possible. Calculate the exact length of the access.
2. Justify that your choice is optimal among all the points of the road, by comparing the distances using a right triangle. A single observation on a drawing is not enough.

Exercise 7.6 · Parameter and set

For a real parameter k, define $\mathcal{E}_k$ as the set of points $M(x, y)$ whose coordinates satisfy

$$x^2 + y^2 - 2x + 4y + k = 0.$$

1. Determine the nature of $\mathcal{E}_k$ according to the value of k. Organise the answer in disjoint cases.
2. When this set is a circle, state its centre and radius; in the other cases, describe exactly the set obtained.

Exercise 7.7 · A parabola as a locus

Optional extension. In an orthonormal frame, fix the point $F(0, 2)$. We look for the points $M(x, y)$ whose distance to F is equal to the distance to the x-axis.

1. Express the square of each of the two distances in terms of x and y.
2. Deduce an equation of the set sought. Check the converse in order to justify that this equation describes exactly all the points sought.

Exercise 7.8 · Delayed review

In an orthonormal frame, $\mathcal{C}$ is the circle with diameter $[AB]$, with $A(0, 0)$ and $B(4, 0)$. Consider the line d with equation $x = 5$.

1. Determine whether $\mathcal{C}$ and d have a common point, by solving an equation.
2. Confirm the answer by comparing the distance from the centre of the circle to the line with the radius of the circle.

Homework. Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 18 p. 279, questions a, b, c and d. Read a nonzero normal vector for each equation, without solving for y.

CQFD, exercise 21 p. 279, questions a and b. Write the condition of orthogonality to the normal, then check that the given point belongs to the line.